

Suppose the domain is U . If the value of $P(c_1, c_2, c_3, \dots, c_n)$ is true for every choice of arguments selected from U , then P is said to be valid in the domain U . If the values $P(c_1, c_2, c_3, \dots, c_n)$ is true for some (but not for all) choices of argument selected from U , then P is said to be satisfiable in the domain U and the values $c_1, c_2, c_3, \dots, c_n$ which make $P(c_1, c_2, c_3, \dots, c_n)$ true are said to satisfy P . If P is not satisfiable in the domain U , then we say P is unsatisfiable in U .

• An expression involving predicate variables and quantifiers and connectives included in the proper manner is called a formula (or well-formed formula or wff) of the first order logic. For example $\forall x P(x) \vee \forall x Q(x)$ is a wff. When we assign particular predicates to P and Q it is called an interpretation.

• A wff involving predicate variables is valid if it is true for every domain no matter how the predicate variables are interpreted.

• A wff is said to be satisfiable if there exists a domain and some interpretation of the predicate variables which makes it true.

• If a wff is not true for any domain or interpretation it is unsatisfiable.

• For example, $\forall x P(x) \rightarrow \exists x P(x)$ is true for all domains and interpretations and hence a valid wff.

• $\exists x P(x) \rightarrow \exists x Q(x)$ may be true for some interpretation but may not be true for some other interpretation.

• Given a wff of first order logic find if it is valid or not. This is called the validity problem of the first-order logic. This is an undecidable problem. (Decidability is discussed in Chapter 13).

• A formal system has axioms and rules of inferences explicitly stated. A theorem is an assertion which can be derived from the axioms using the rules of inference.

An argument or ~~formal~~ proof is a step-by-step derivation of the theorem from axioms using rules of inferences.